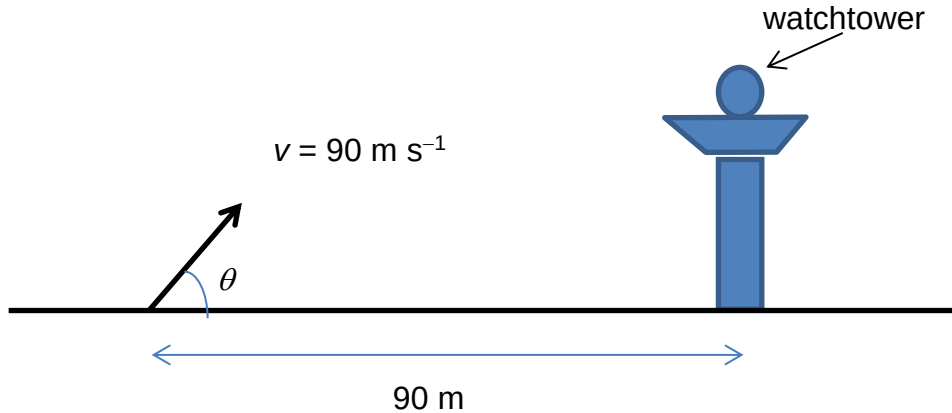


- 3 A projectile is fired at a speed of 90 m s^{-1} at sea level at an angle of $\theta = 31.6^\circ$ with respect to the horizontal as shown below. The projectile hits the top of a watchtower located 90 m away.



How high is the top of the watchtower above sea level? Neglect air resistance.

- A 49 m
- B 72 m
- C 83 m
- D 113 m

A Horizontal displacement of cannon ball $S_x = 90 \cos 31.6^\circ \times t = 90$
 $\Rightarrow t = 1.174 \text{ s}$
Vertical displacement of cannon ball $S_y = 90 \sin 31.6 (1.174) - \frac{1}{2} (9.81) (1.174)^2$
 $= 48.6 \text{ m}$
 $= 49 \text{ m} = \text{height of the top of the watchtower above sea level}$

- 4 Two blocks X and Y of masses m and $2m$ respectively, are accelerated along a rough